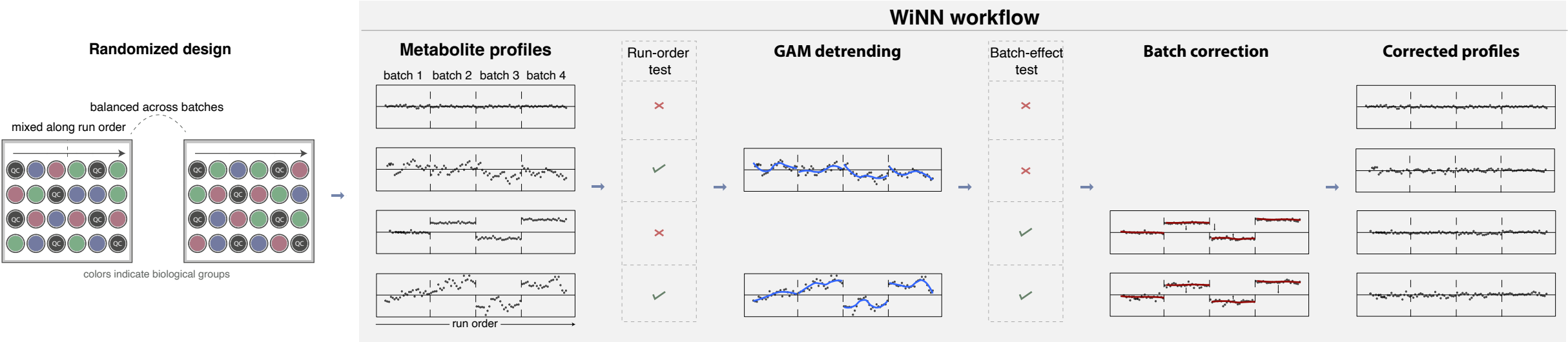
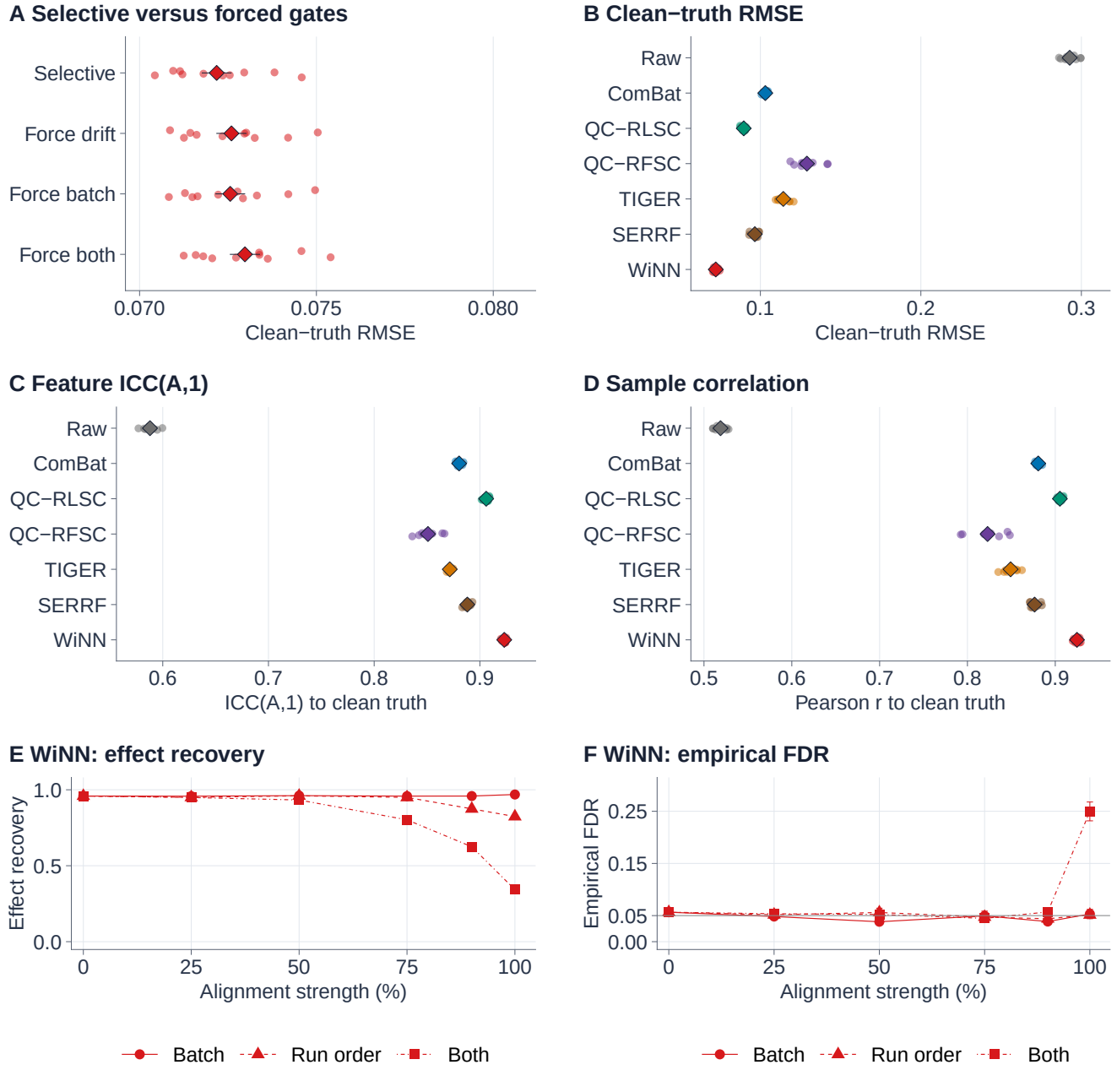
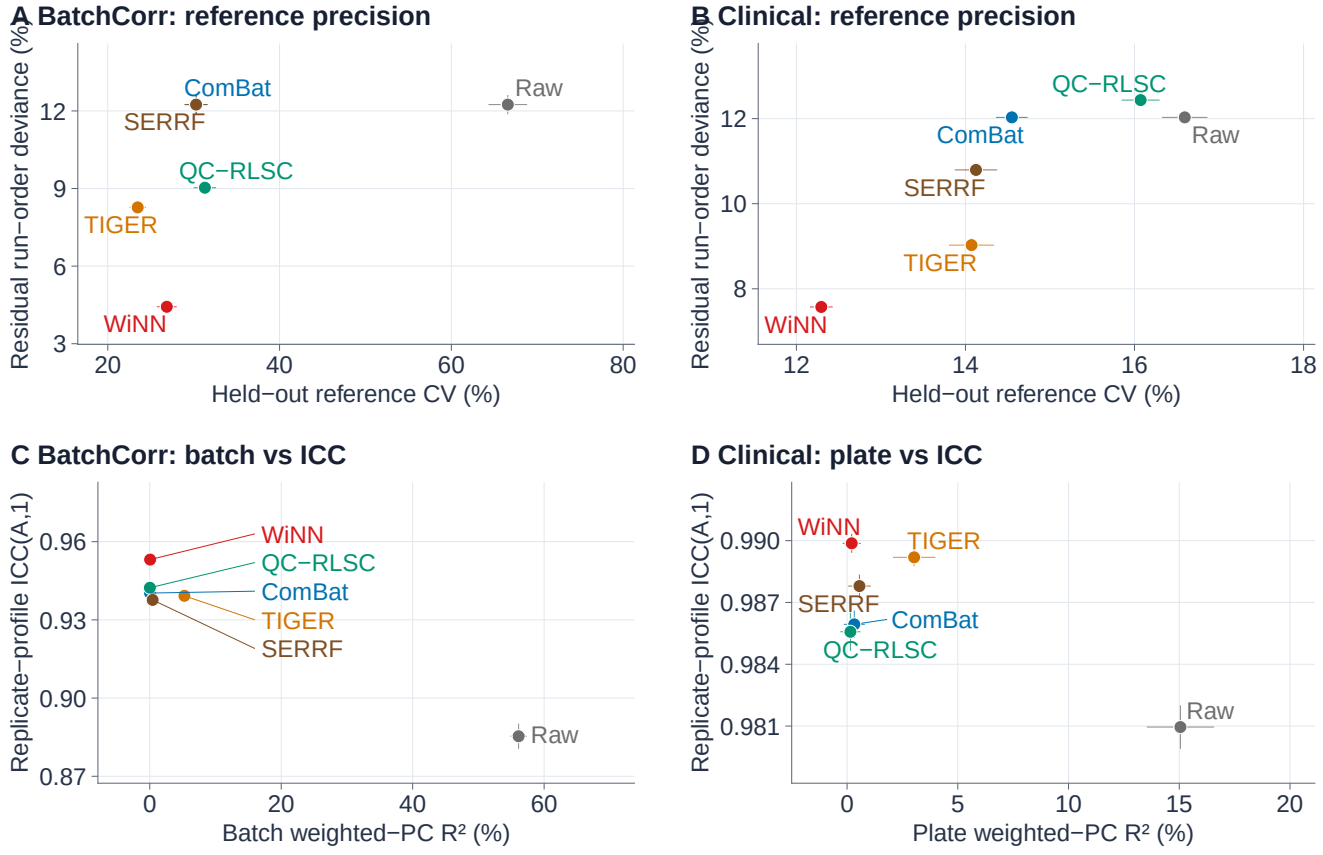
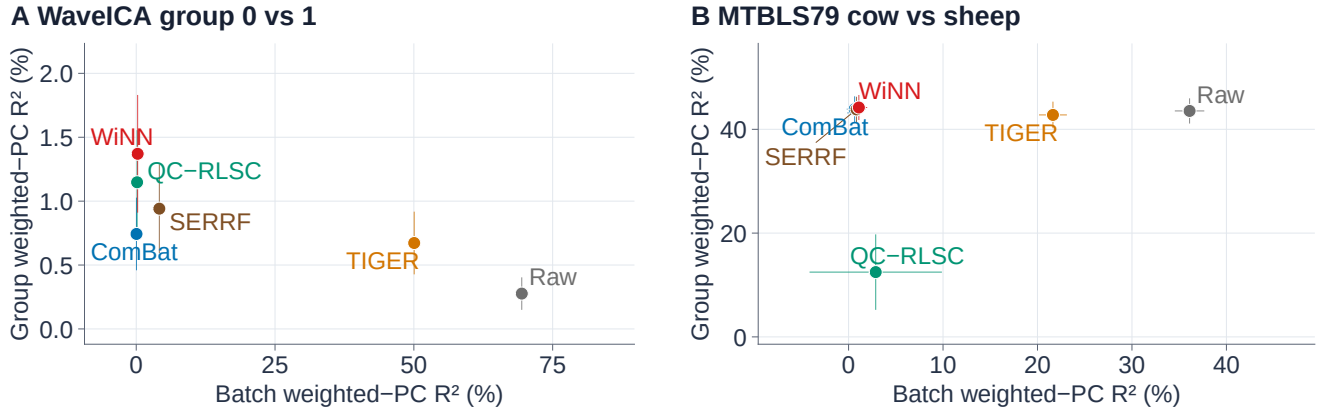
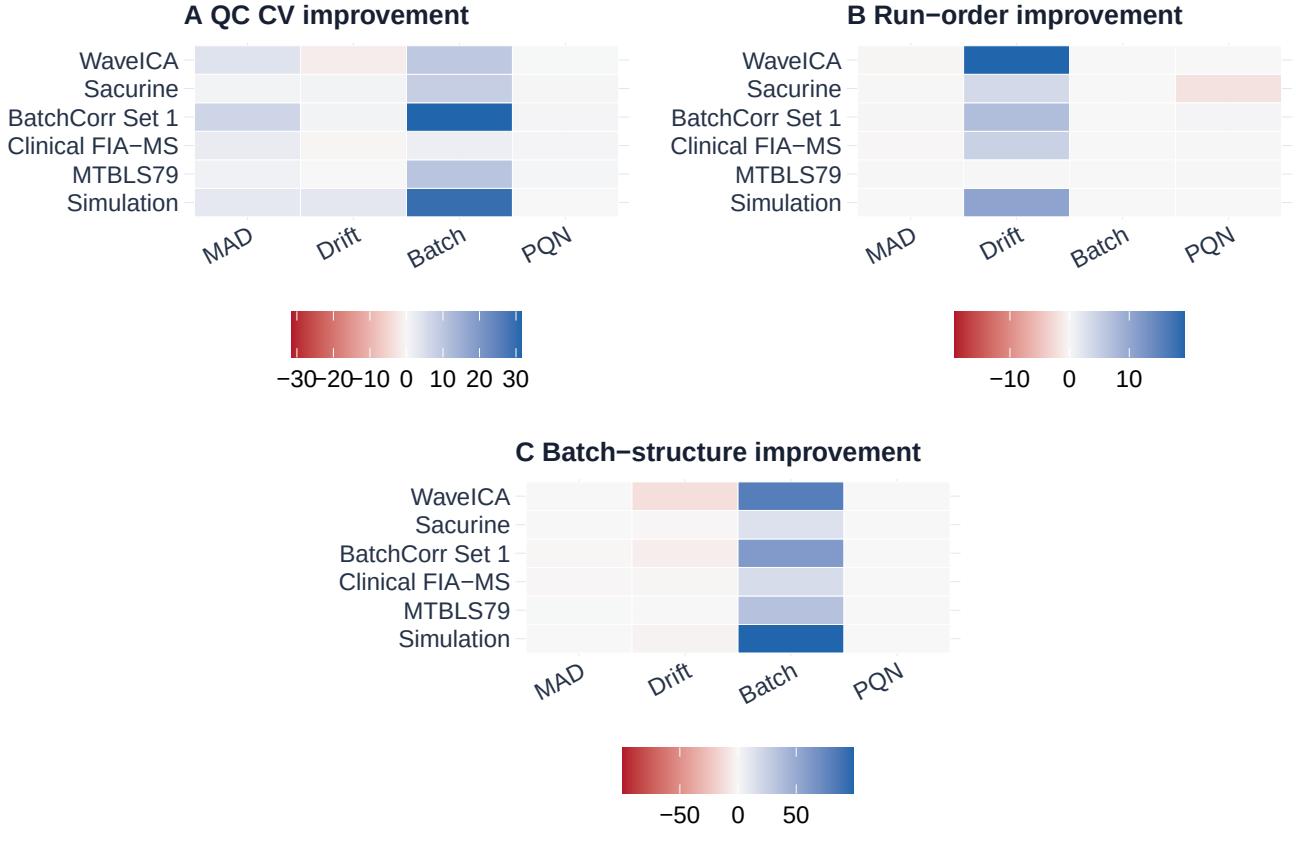
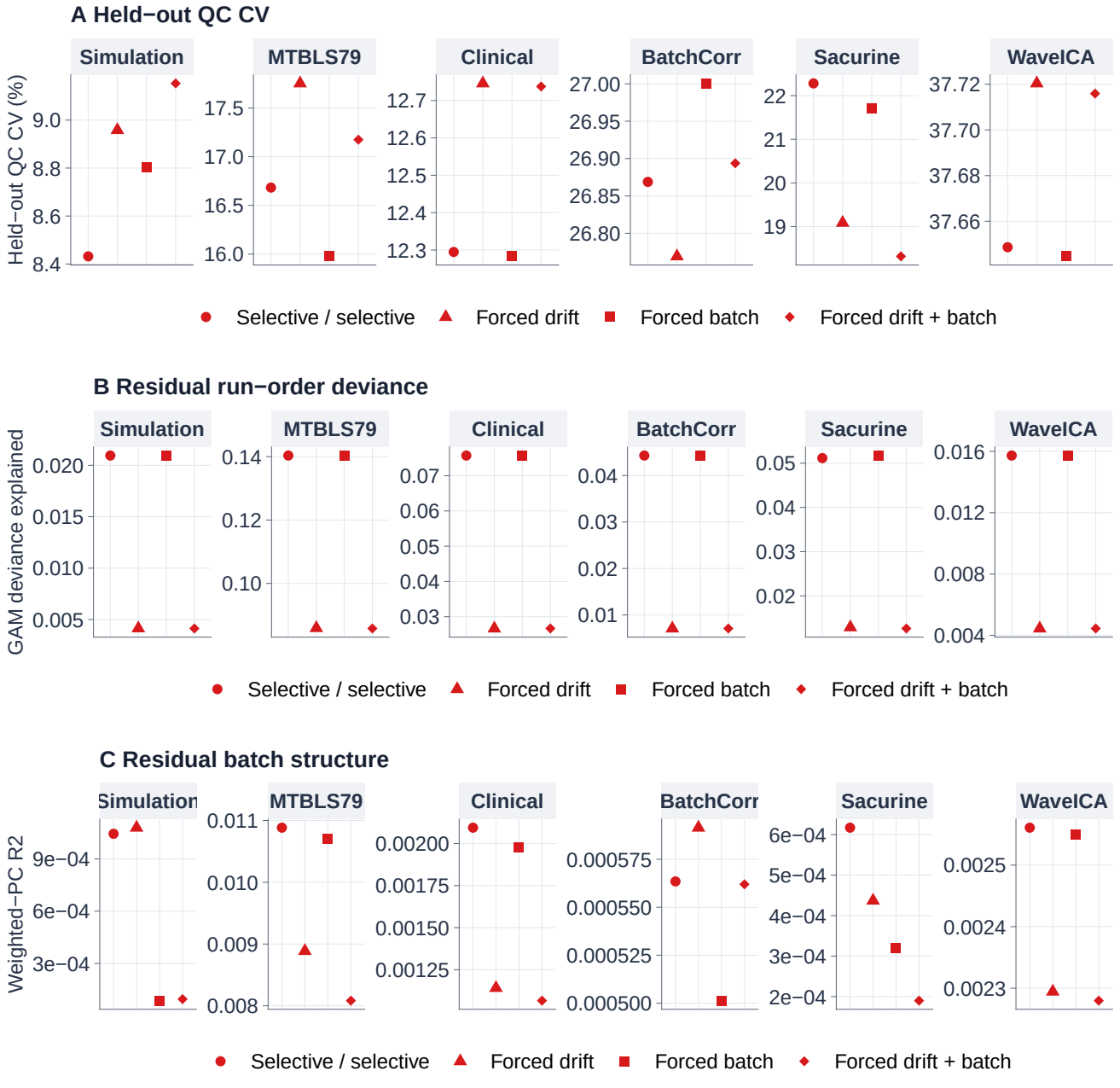

Randomized design
balanced across batches
mixed along run order
QC
colors indicate biological groups
WiNN workflow
Metabolite profiles
batch 1
batch 2
batch 3
batch 4
run order
Run-order test
GAM detrending
Batch-effect test
Batch correction
Corrected profiles

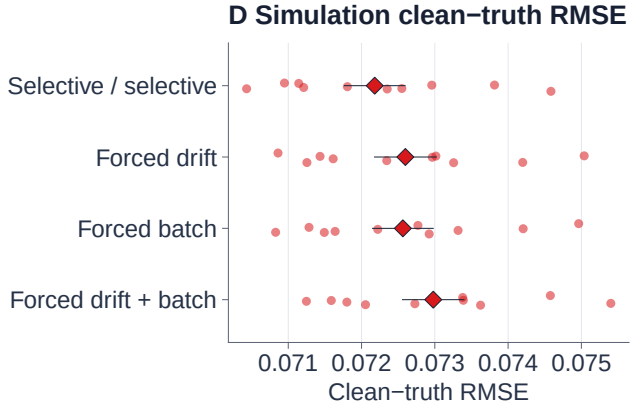

A Selective versus forced gates
Selective
Force drift
Force batch
Force both
0.070
0.075
0.080
Clean−truth RMSE
B Clean−truth RMSE
Raw
ComBat
QC−RLSC
QC−RFSC
TIGER
SERRF
WiNN
0.1
0.2
0.3
Clean−truth RMSE
C Feature ICC(A,1)
0.6
0.7
0.8
0.9
ICC(A,1) to clean truth
D Sample correlation
0.5
0.6
0.7
0.8
0.9
Pearson r to clean truth
E WiNN: effect recovery
Effect recovery
1.0
0.5
0.0
0
25
50
75
100
Alignment strength (%)
F WiNN: empirical FDR
Empirical FDR
0.25
0.15
0.05
0.00
0
25
50
75
100
Alignment strength (%)
Batch
Run order
Both
Batch
Run order
Both

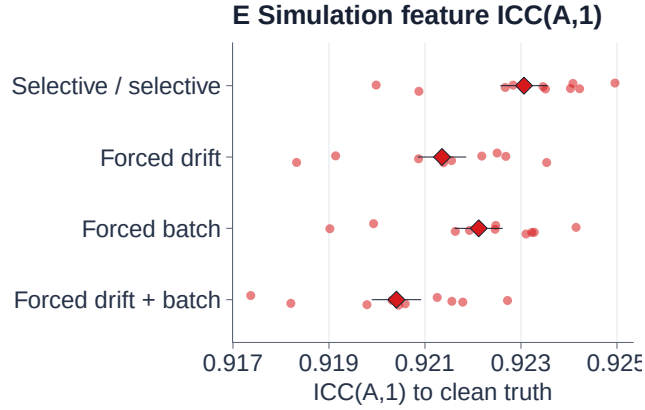

A BatchCorr: reference precision
Residual run−order deviance (%)
Held−out reference CV (%)
ComBat
Raw
SERRF
QC−RLSC
TIGER
WiNN
B Clinical: reference precision
Residual run−order deviance (%)
Held−out reference CV (%)
QC−RLSC
ComBat
Raw
SERRF
TIGER
WiNN
C BatchCorr: batch vs ICC
Replicate−profile ICC(A,1)
Batch weighted−PC R² (%)
WiNN
QC−RLSC
ComBat
TIGER
SERRF
Raw
D Clinical: plate vs ICC
Replicate−profile ICC(A,1)
Plate weighted−PC R² (%)
WiNN
TIGER
SERRF
ComBat
QC−RLSC
Raw

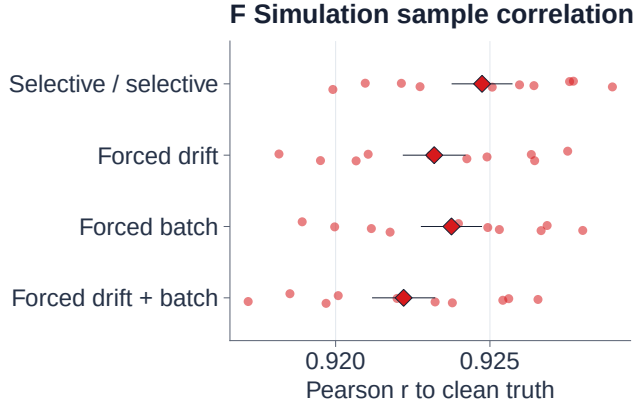

A WaveICA group 0 vs 1
Group weighted−PC R² (%)
Batch weighted−PC R² (%)
2.0
1.5
1.0
0.5
0.0
0
25
50
75
WiNN
QC−RLSC
SERRF
ComBat
TIGER
Raw
B MTBLS79 cow vs sheep
Group weighted−PC R² (%)
Batch weighted−PC R² (%)
40
20
0
0
10
20
30
40
WiNN
ComBat
SERRF
TIGER
Raw
QC−RLSC

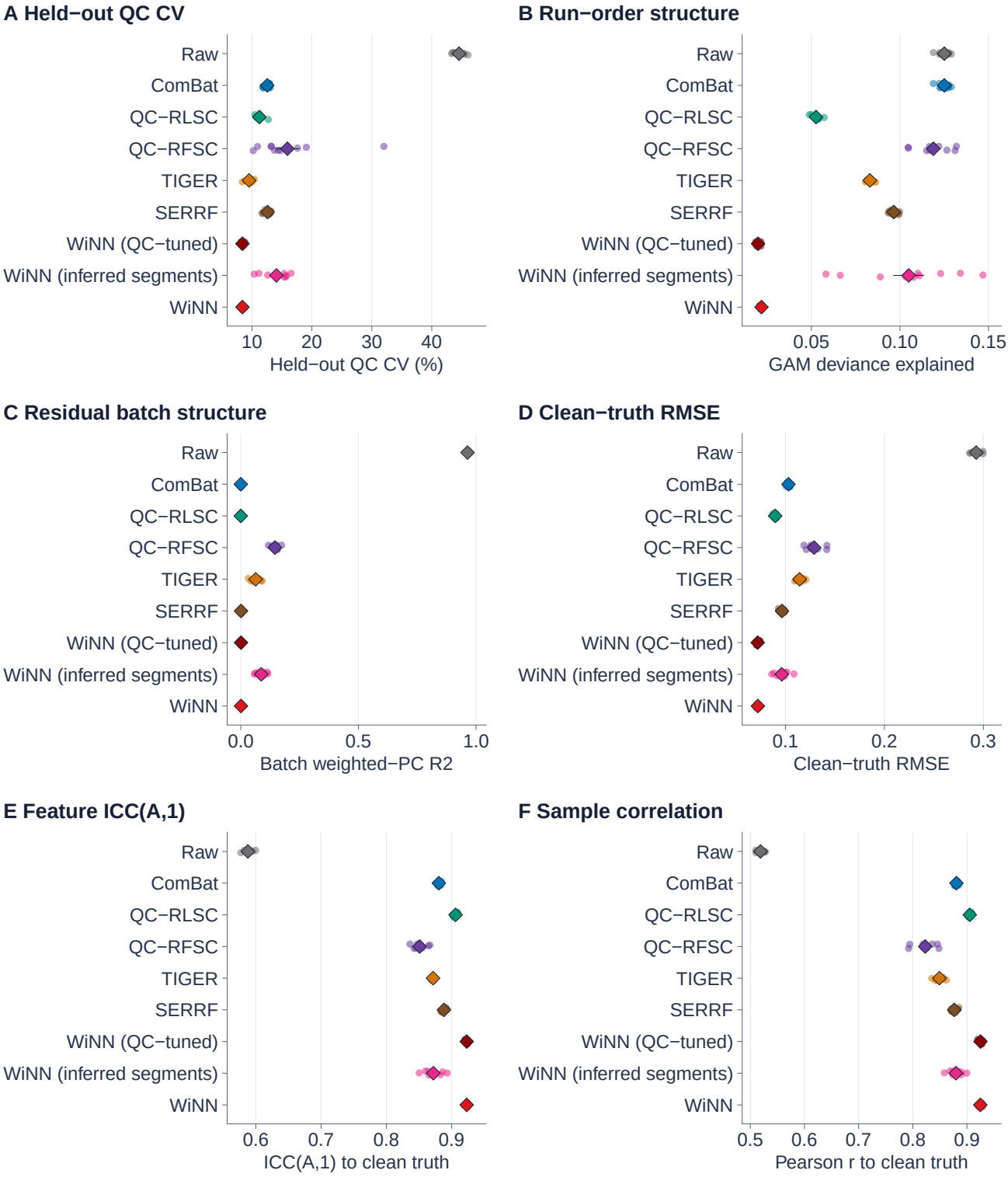

A QC CV improvement
WaveICA
Sacurine
BatchCorr Set 1
Clinical FIA−MS
MTBLS79
Simulation
MAD
Drift
Batch
PQN
−30 −20 −10 0 10 20 30
B Run−order improvement
WaveICA
Sacurine
BatchCorr Set 1
Clinical FIA−MS
MTBLS79
Simulation
MAD
Drift
Batch
PQN
−10 0 10
C Batch−structure improvement
WaveICA
Sacurine
BatchCorr Set 1
Clinical FIA−MS
MTBLS79
Simulation
MAD
Drift
Batch
PQN
−50 0 50

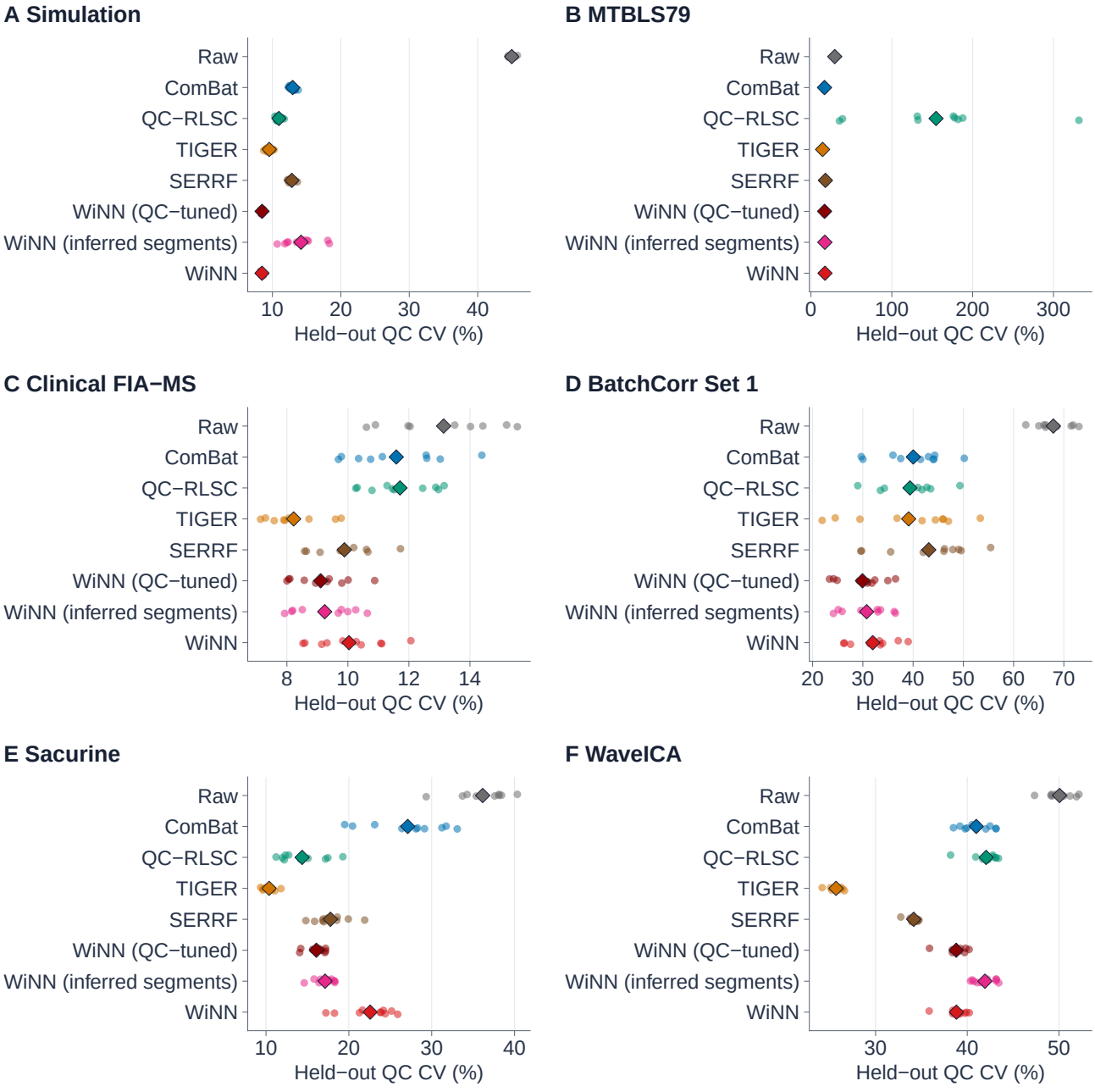

A Held−out QC CV
Held−out QC CV (%)
Simulation
MTBLS79
Clinical
BatchCorr
Sacurine
WaveICA
B Residual run−order deviance
GAM deviance explained
C Residual batch structure
Weighted−PC R2
Selective / selective
Forced drift
Forced batch
Forced drift + batch

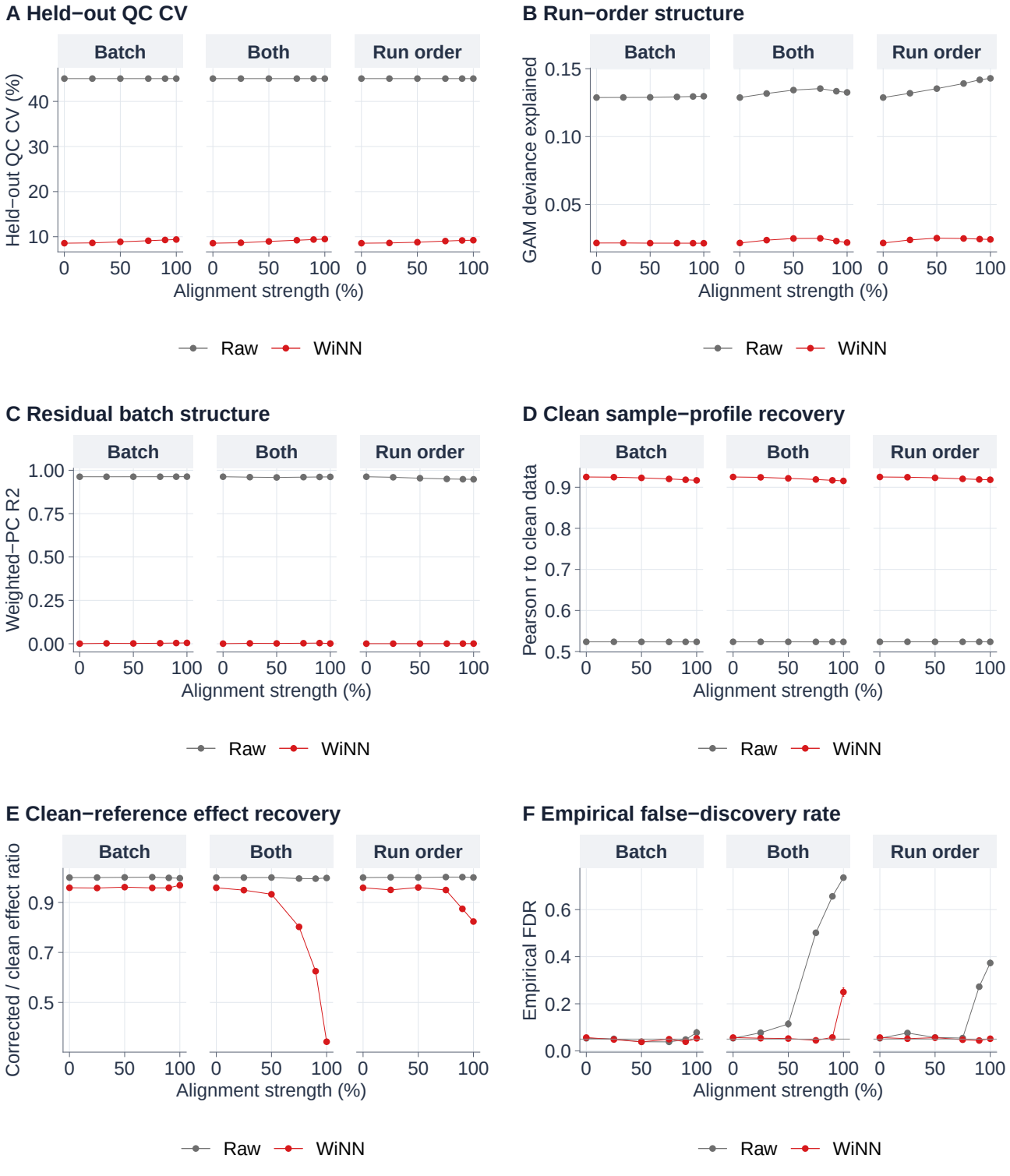

D Simulation clean−truth RMSE
Selective / selective
Forced drift
Forced batch
Forced drift + batch
0.071
0.072
0.073
0.074
0.075
Clean−truth RMSE


E Simulation feature ICC(A,1)
Selective / selective
Forced drift
Forced batch
Forced drift + batch
0.917
0.919
0.921
0.923
0.925
ICC(A,1) to clean truth


F Simulation sample correlation
Selective / selective
Forced drift
Forced batch
Forced drift + batch
0.920
0.925
Pearson r to clean truth

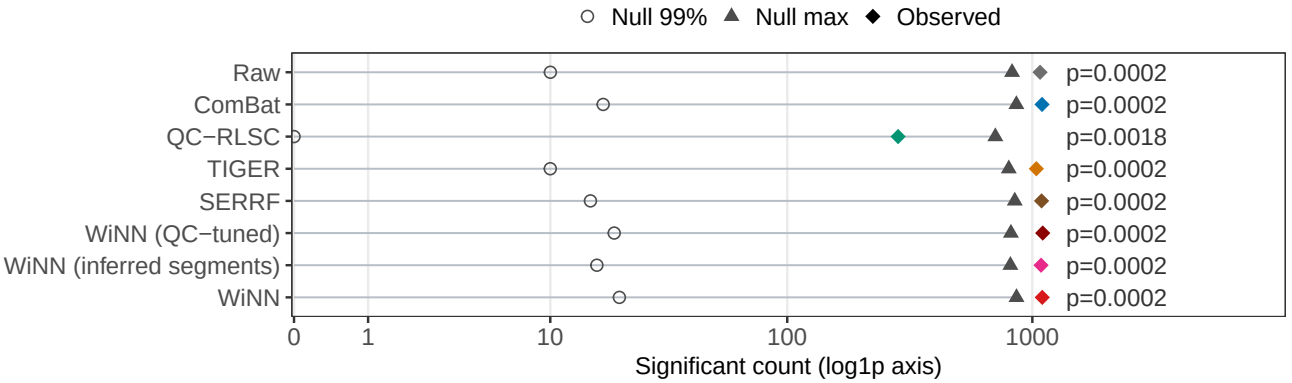

A Held−out QC CV
Raw
ComBat
QC−RLSC
QC−RFSC
TIGER
SERRF
WiNN (QC−tuned)
WiNN (inferred segments)
WiNN
10
20
30
40
Held−out QC CV (%)
B Run−order structure
0.05
0.10
0.15
GAM deviance explained
C Residual batch structure
0.0
0.5
1.0
Batch weighted−PC R2
D Clean−truth RMSE
0.1
0.2
0.3
Clean−truth RMSE
E Feature ICC(A,1)
0.6
0.7
0.8
0.9
ICC(A,1) to clean truth
F Sample correlation
0.5
0.6
0.7
0.8
0.9
Pearson r to clean truth


A Simulation
B MTBLS79
C Clinical FIA−MS
D BatchCorr Set 1
E Sacurine
F WaveICA
Raw
ComBat
QC−RLSC
TIGER
SERRF
WiNN (QC−tuned)
WiNN (inferred segments)
WiNN
Held−out QC CV (%)


A Held−out QC CV
Batch
Both
Run order
Held−out QC CV (%)
40
30
20
10
0
50
100
Alignment strength (%)
Raw
WiNN
B Run−order structure
GAM deviance explained
0.15
0.10
0.05
C Residual batch structure
Weighted−PC R2
1.00
0.75
0.50
0.25
0.00
D Clean sample−profile recovery
Pearson r to clean data
0.9
0.8
0.7
0.6
0.5
E Clean−reference effect recovery
Corrected / clean effect ratio
F Empirical false−discovery rate
Empirical FDR
0.6
0.4
0.2
0.0

A MTBLS79 cow vs sheep

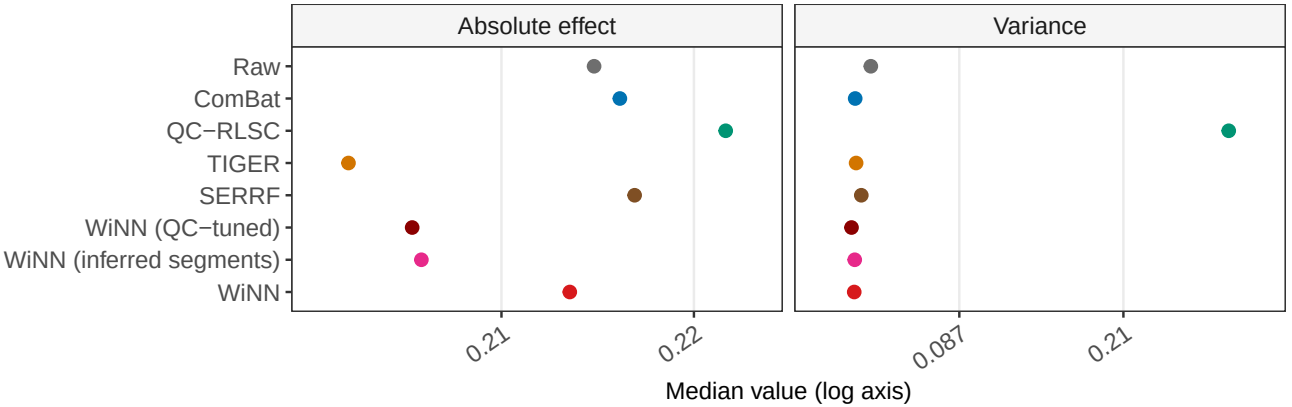
Null tails and observed counts



5,000 constrained permutations.
p = (1 + null counts >= observed) / 5,001.

B MTBLS79 cow vs sheep

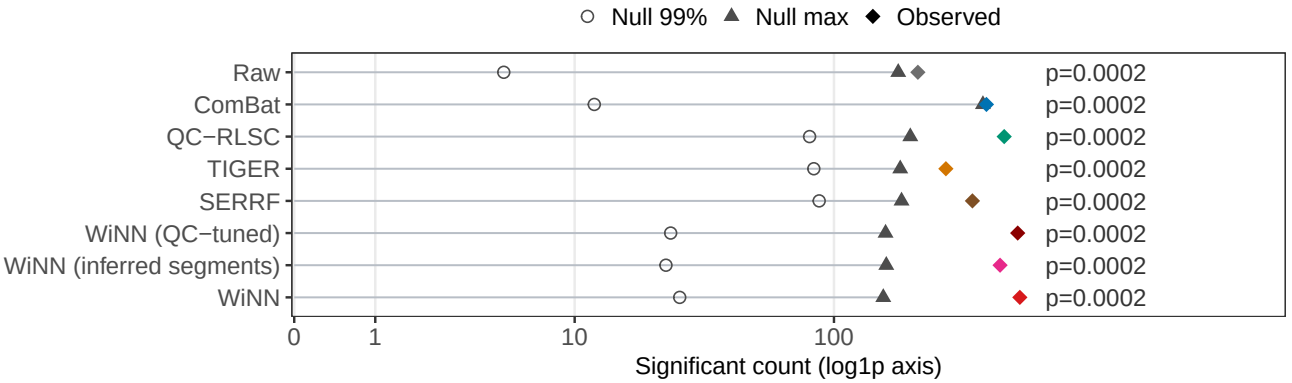
Absolute effects and variances



Effects: log1p intensity scale.

C WaveICA deposited groups

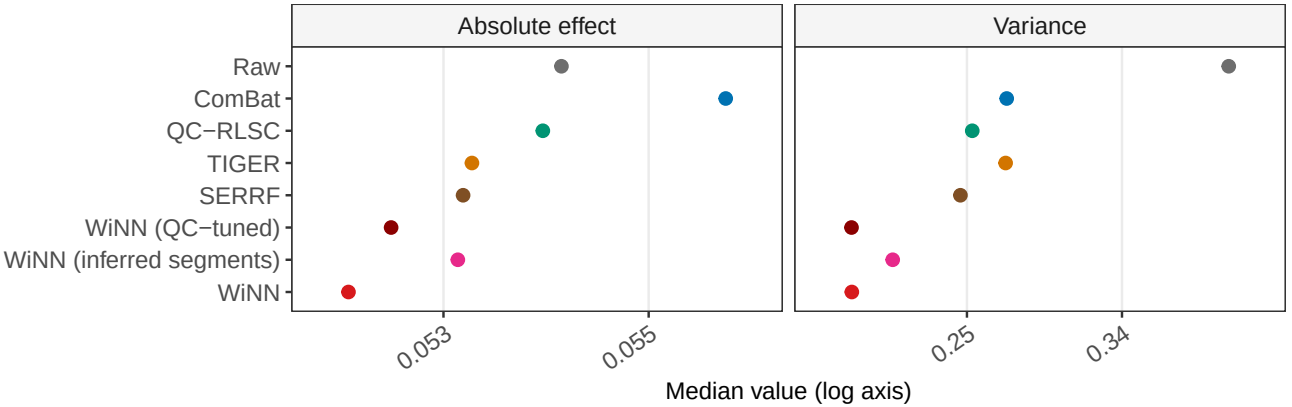
Null tails and observed counts



5,000 constrained permutations.
p = (1 + null counts >= observed) / 5,001.

D WaveICA deposited groups

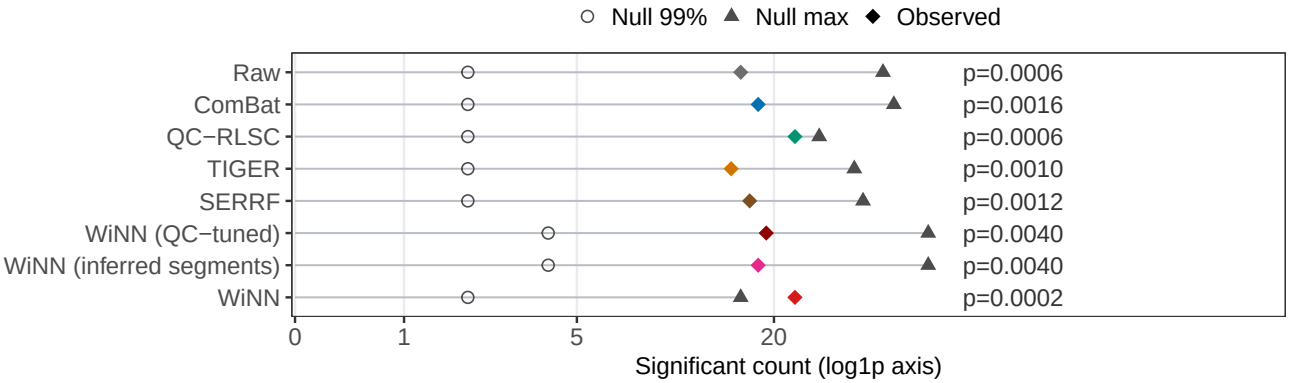
Absolute effects and variances



Effects: log1p intensity scale.

E Sacurine age

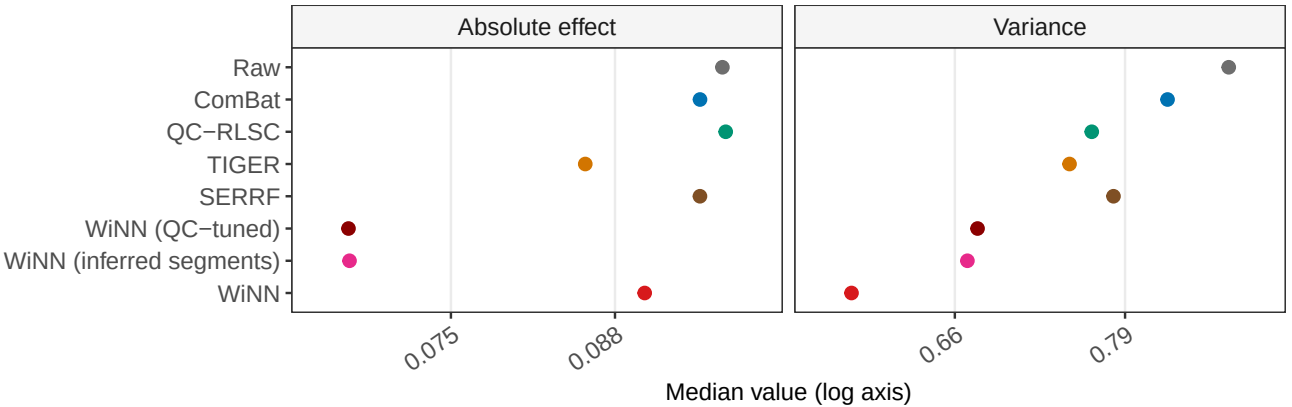
Null tails and observed counts



5,000 constrained permutations.
p = (1 + null counts >= observed) / 5,001.

F Sacurine age

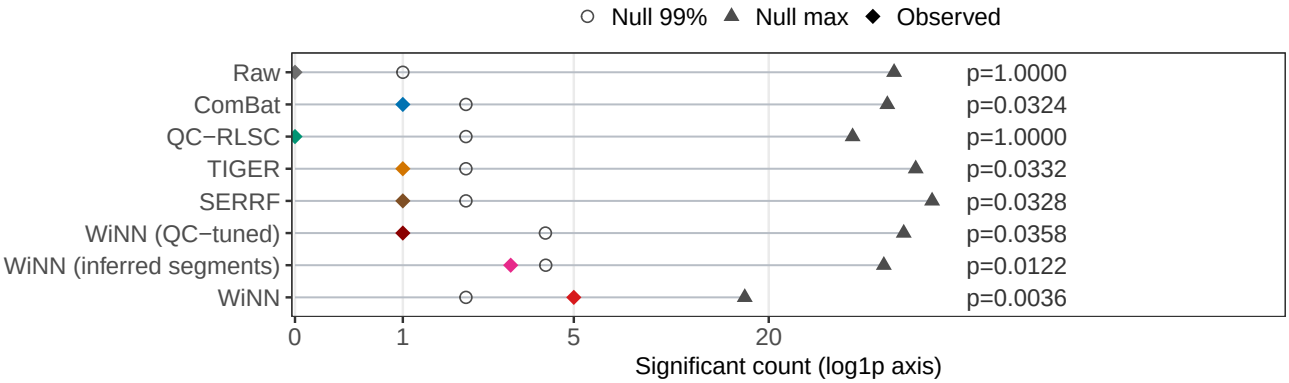
Absolute effects and variances



Effects: log1p after osmolality scaling.

G Sacurine BMI

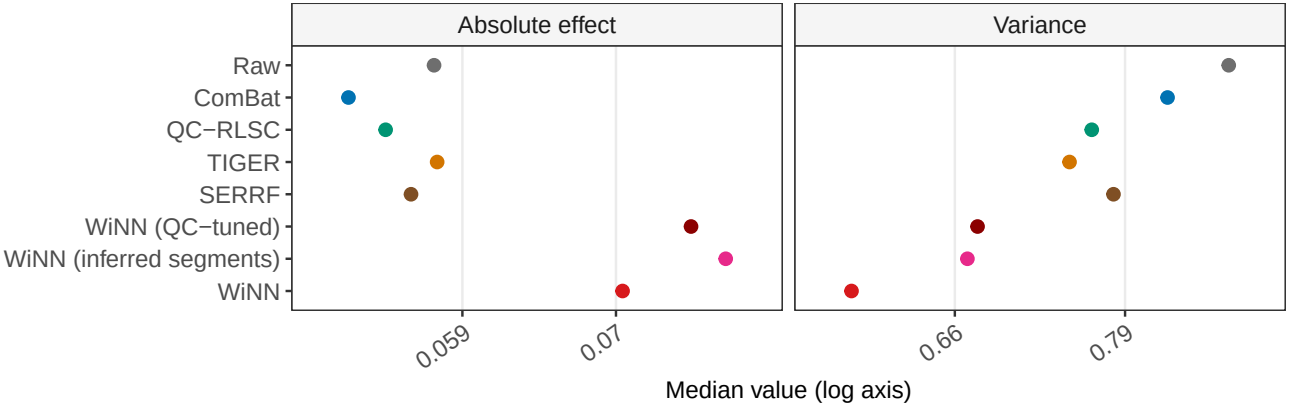
Null tails and observed counts



5,000 constrained permutations.
p = (1 + null counts >= observed) / 5,001.

H Sacurine BMI

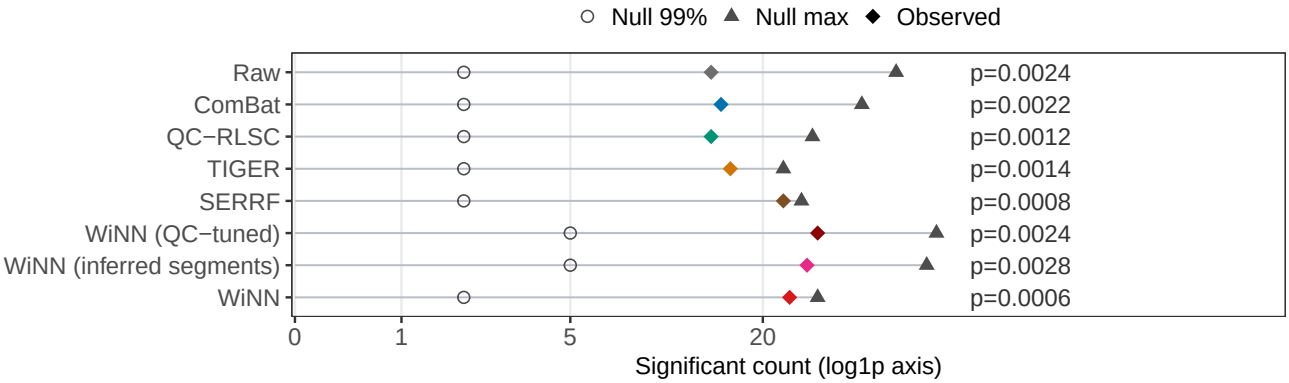
Absolute effects and variances



Effects: log1p after osmolality scaling.

I Sacurine gender

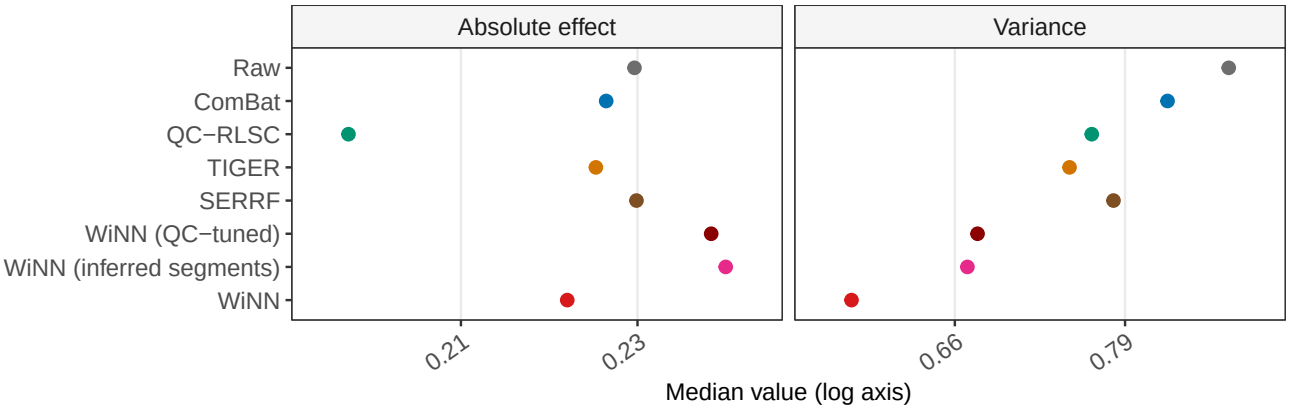
Null tails and observed counts



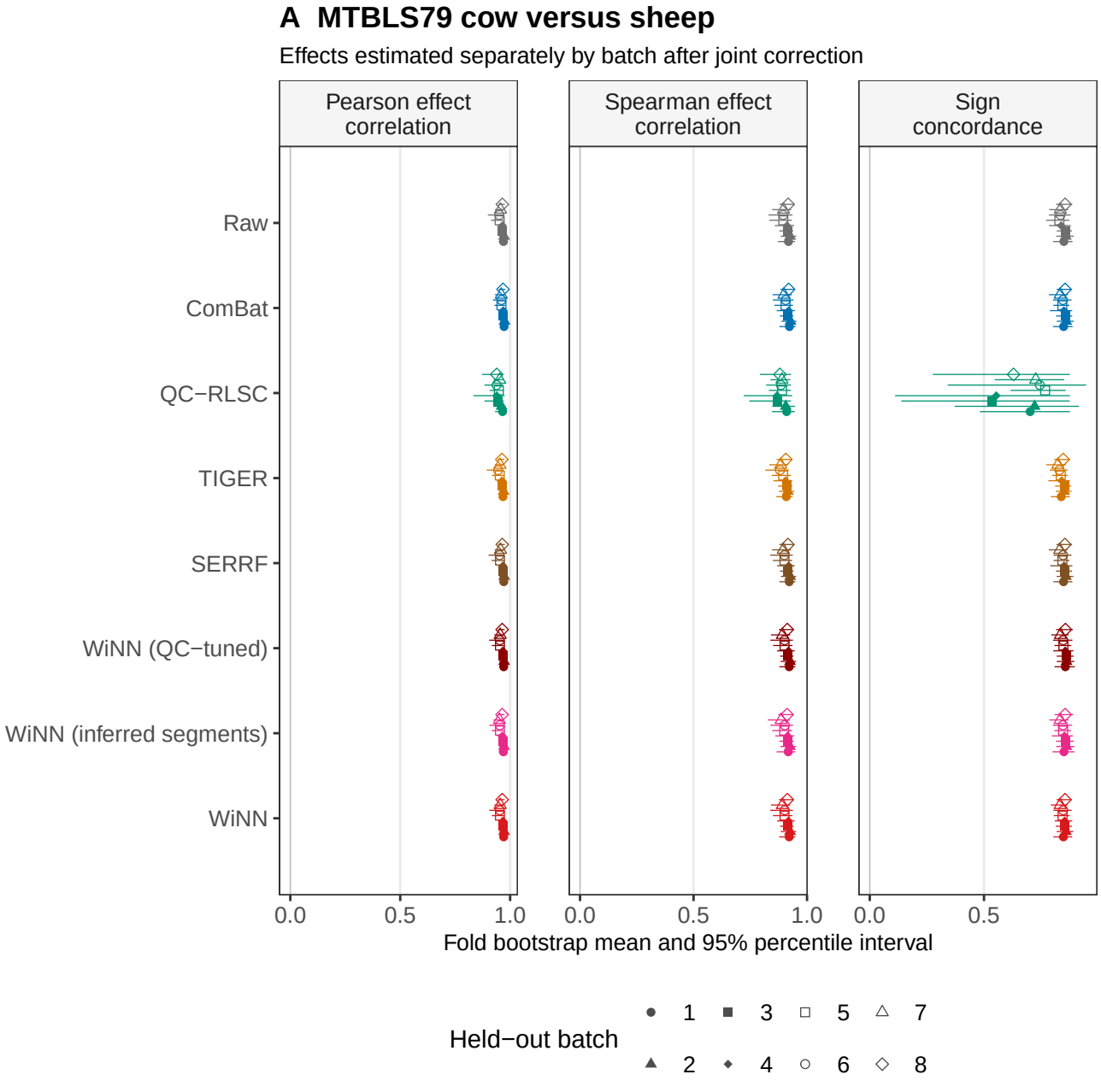
5,000 constrained permutations.
p = (1 + null counts >= observed) / 5,001.

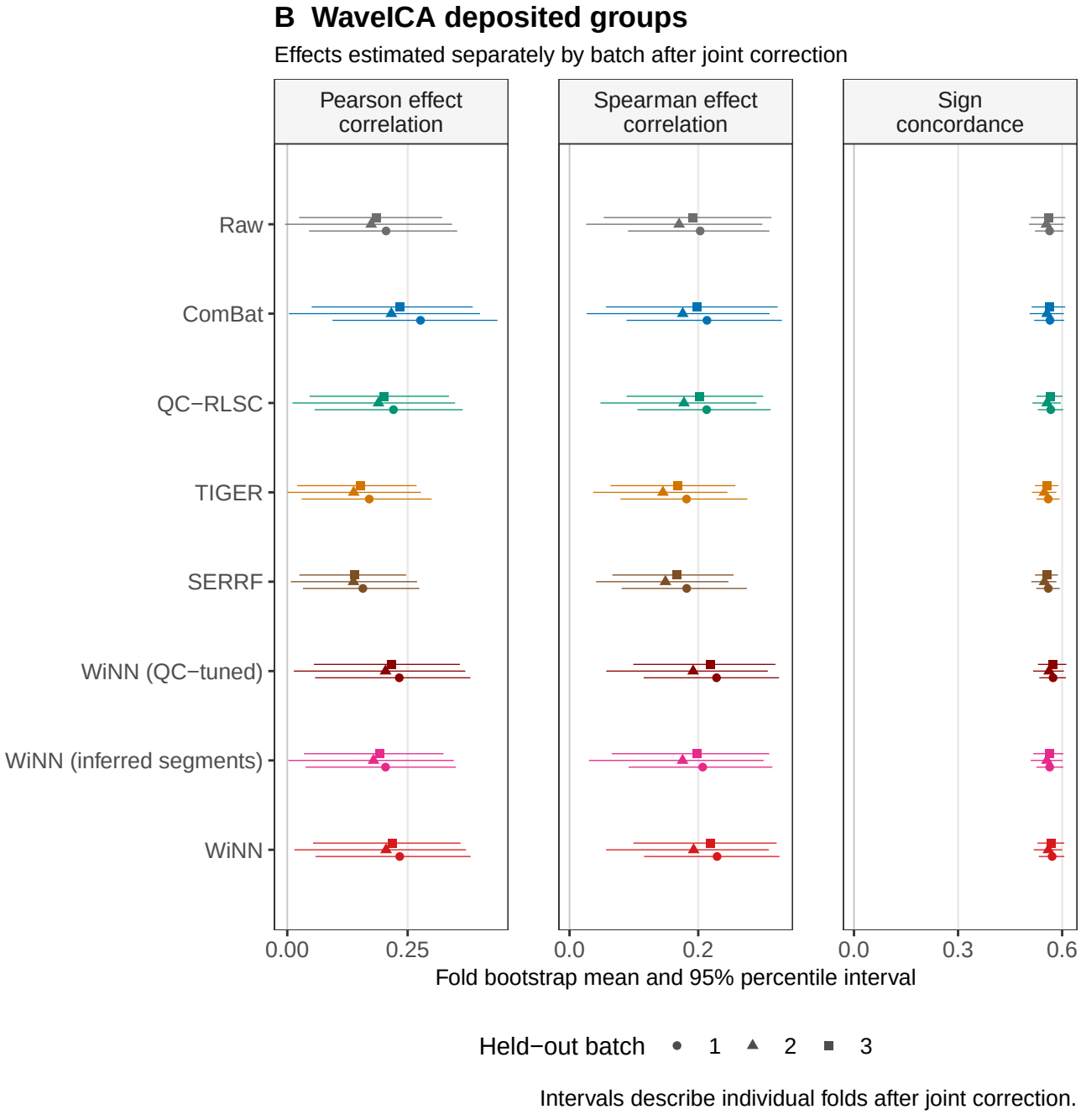
J Sacurine gender

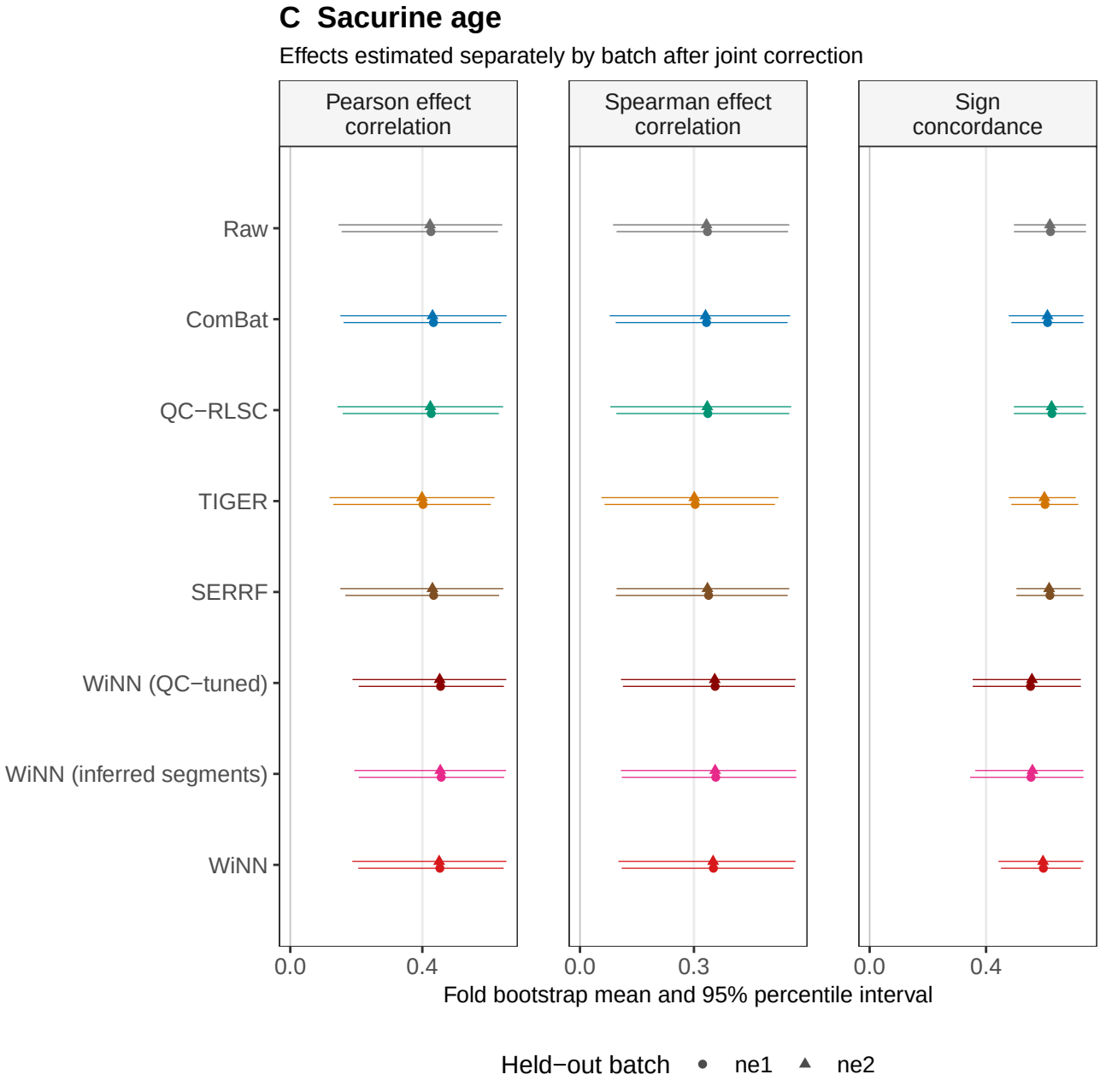
Absolute effects and variances

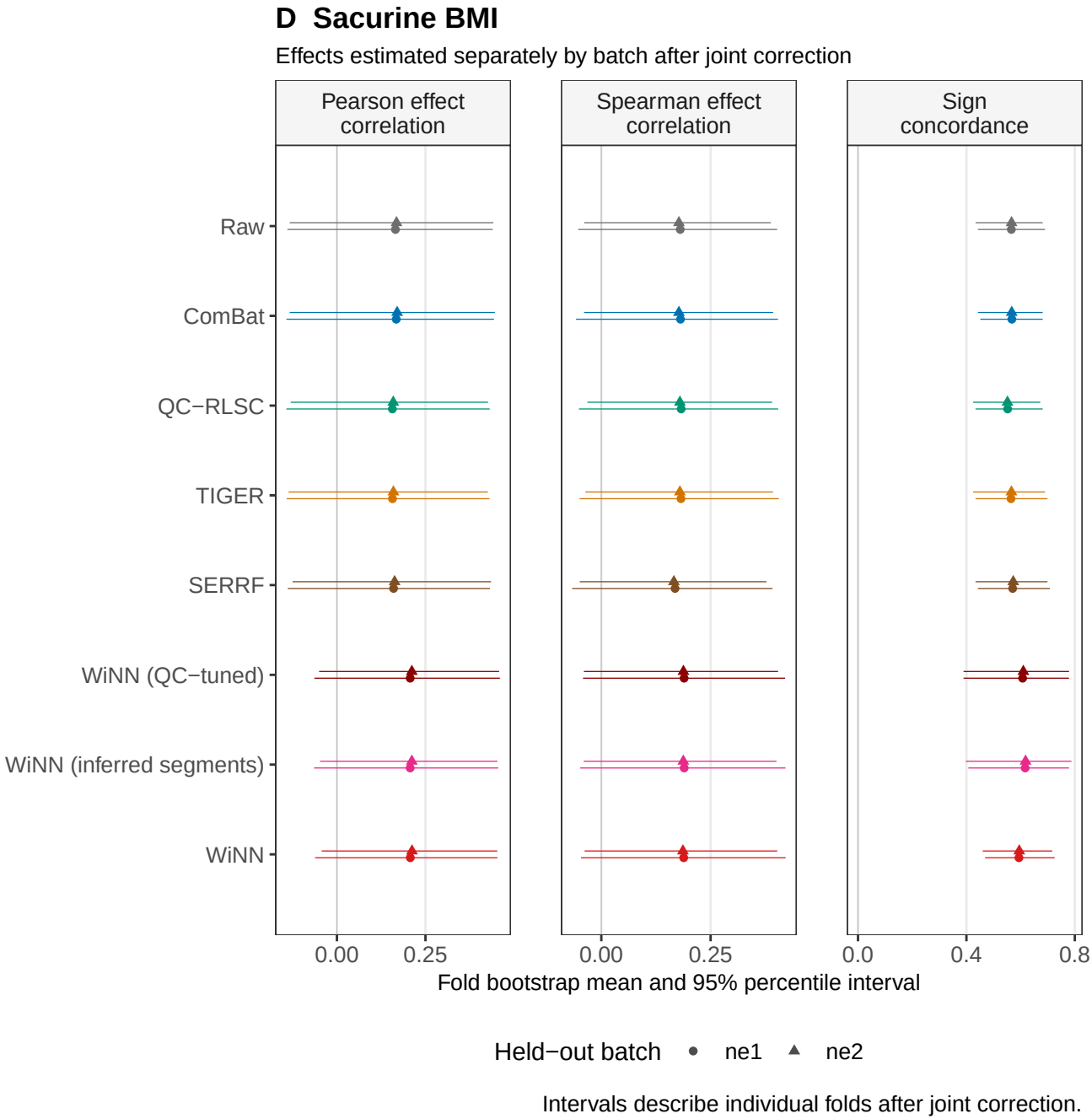


Effects: log1p after osmolality scaling.


A MTBLS79 cow versus sheep
Effects estimated separately by batch after joint correction
Pearson effect correlation
Spearman effect correlation
Sign concordance
Raw
ComBat
QC−RLSC
TIGER
SERRF
WiNN (QC−tuned)
WiNN (inferred segments)
WiNN
0.0
0.5
1.0
Fold bootstrap mean and 95% percentile interval
Held−out batch
1
2
3
4
5
6
7
8
Intervals describe individual folds after joint correction.

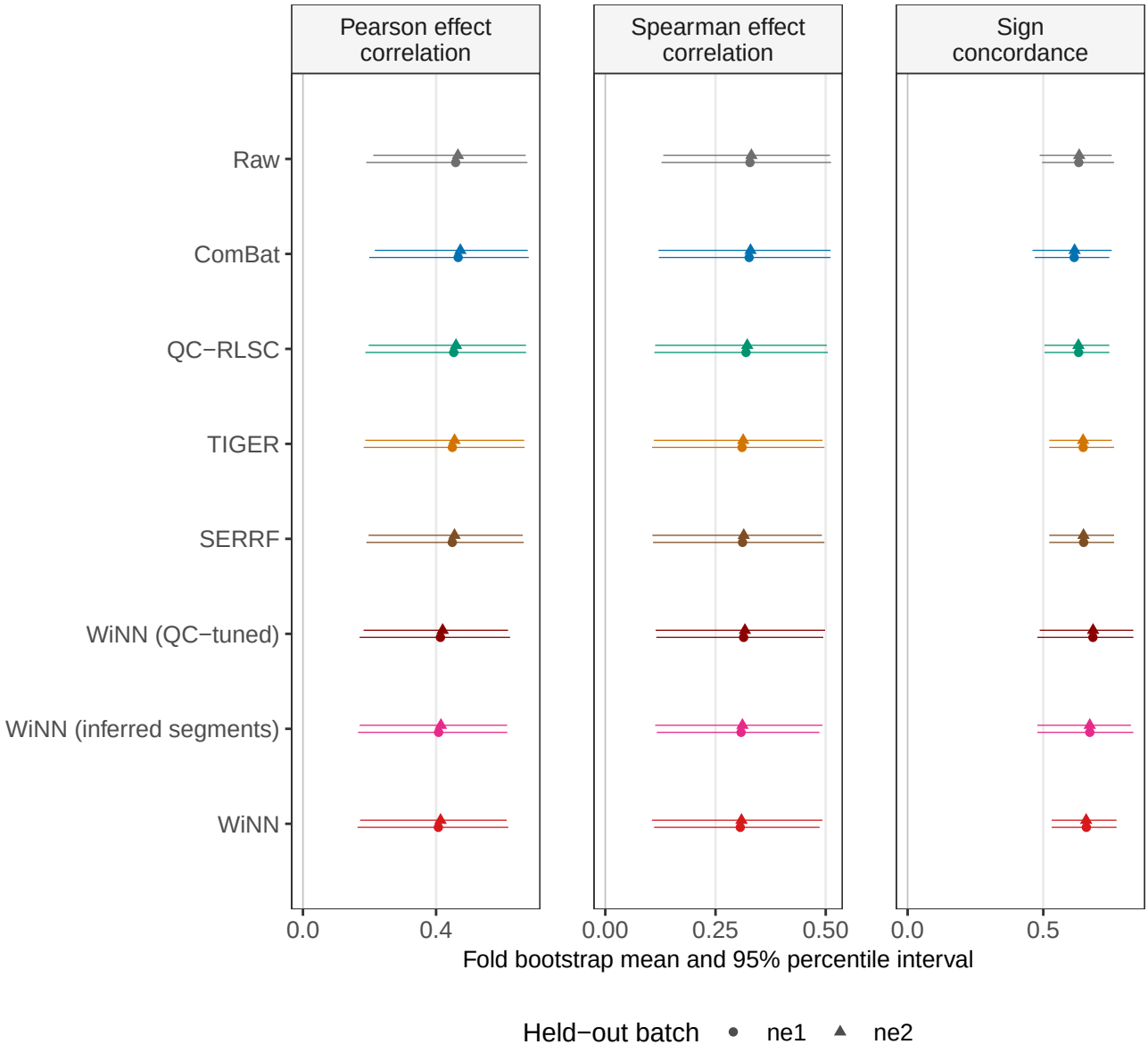

B WaveICA deposited groups
Effects estimated separately by batch after joint correction
Pearson effect correlation
Spearman effect correlation
Sign concordance
Raw
ComBat
QC−RLSC
TIGER
SERRF
WiNN (QC−tuned)
WiNN (inferred segments)
WiNN
0.00
0.25
0.0
0.2
0.0
0.3
0.6
Fold bootstrap mean and 95% percentile interval
Held−out batch
1
2
3
Intervals describe individual folds after joint correction.


C Sacurine age
Effects estimated separately by batch after joint correction
Pearson effect correlation
Spearman effect correlation
Sign concordance
Raw
ComBat
QC−RLSC
TIGER
SERRF
WiNN (QC−tuned)
WiNN (inferred segments)
WiNN
0.0
0.4
0.0
0.3
0.0
0.4
Fold bootstrap mean and 95% percentile interval
Held−out batch
ne1
ne2
Intervals describe individual folds after joint correction.

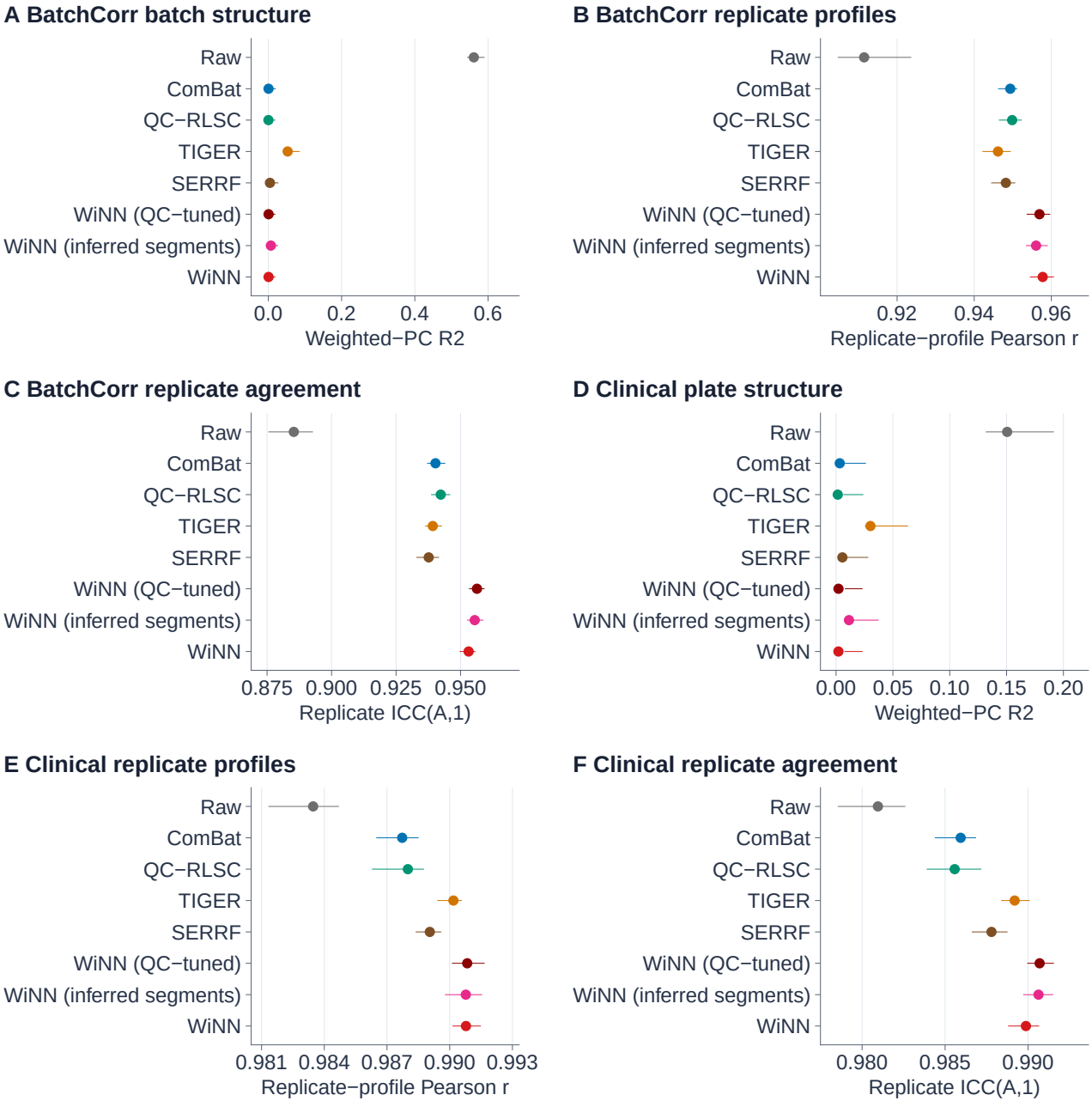

D Sacurine BMI
Effects estimated separately by batch after joint correction
Pearson effect correlation
Spearman effect correlation
Sign concordance
Raw
ComBat
QC−RLSC
TIGER
SERRF
WiNN (QC−tuned)
WiNN (inferred segments)
WiNN
0.00
0.25
0.00
0.25
0.0
0.4
0.8
Fold bootstrap mean and 95% percentile interval
Held−out batch
ne1
ne2
Intervals describe individual folds after joint correction.

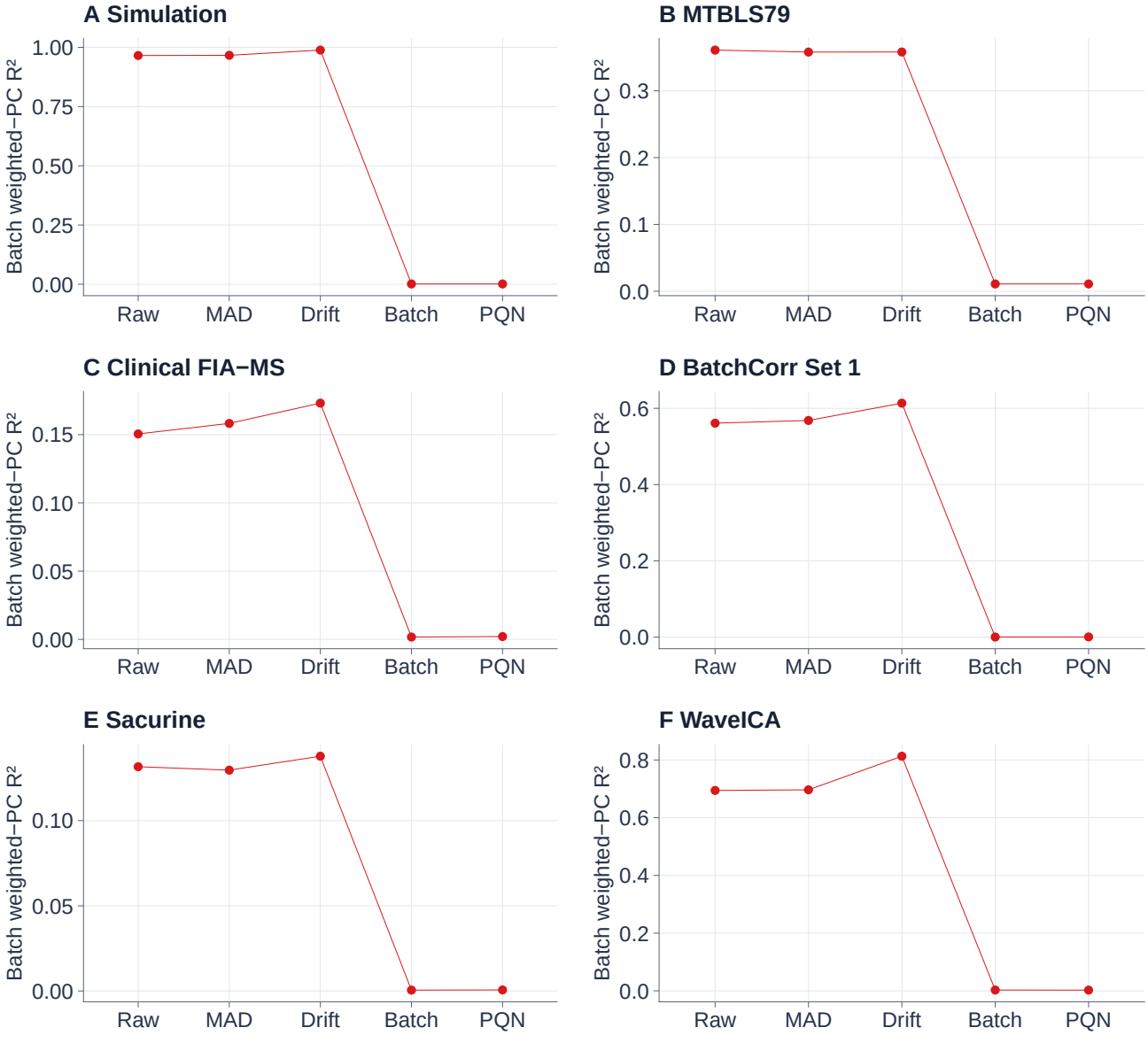
E Sacurine gender

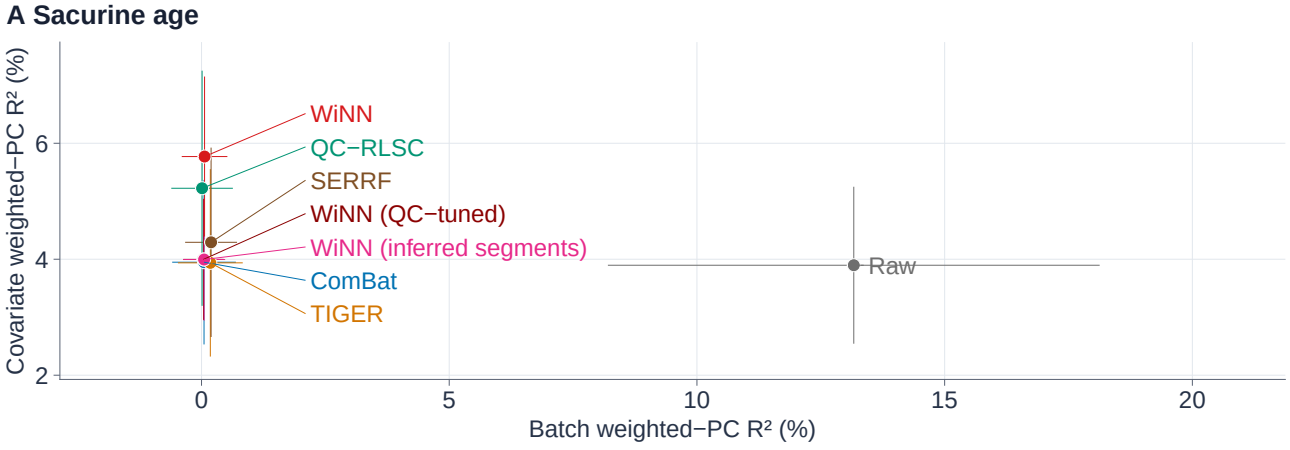
Effects estimated separately by batch after joint correction

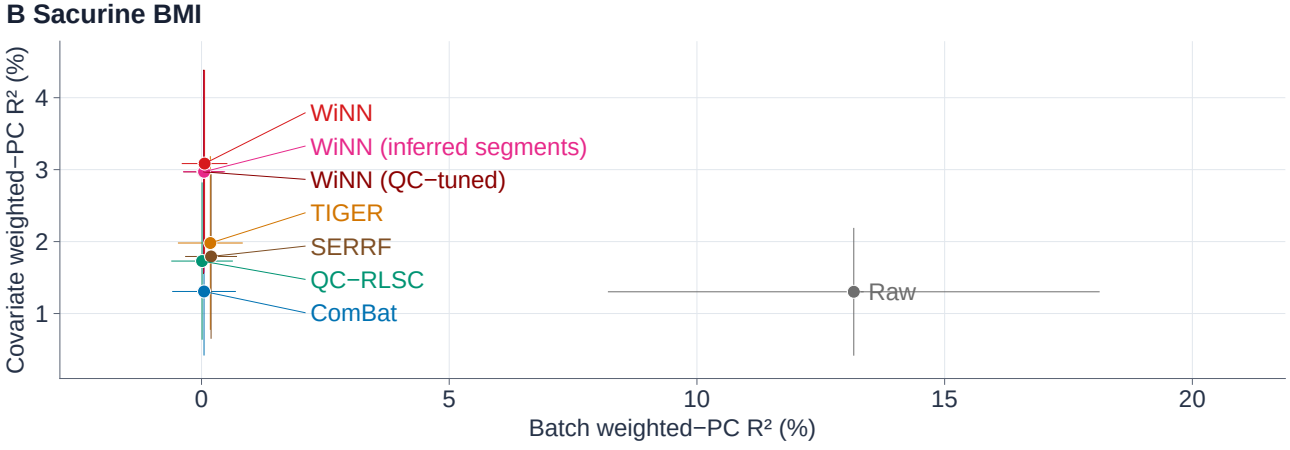


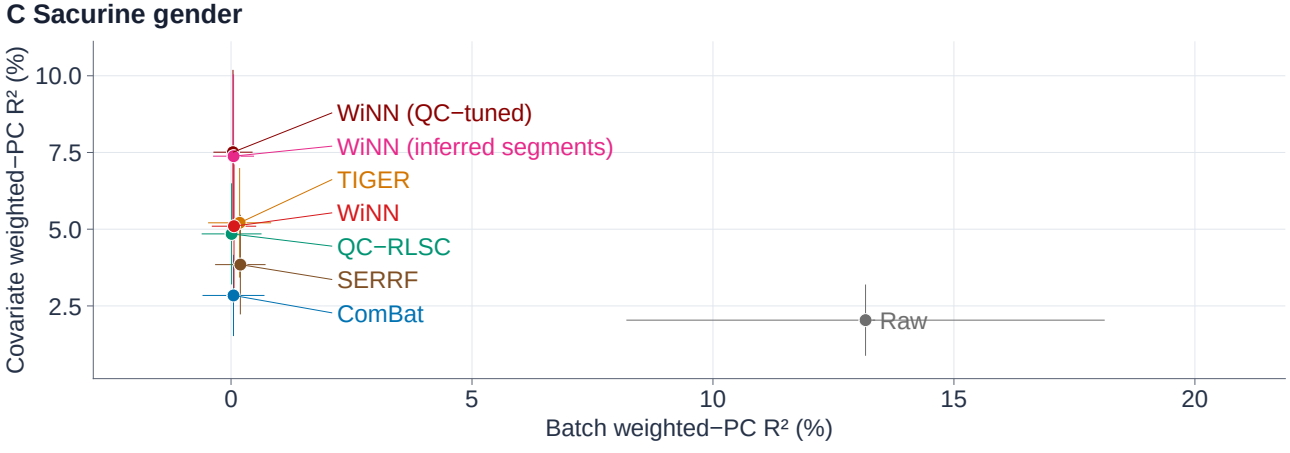
Intervals describe individual folds after joint correction.


A BatchCorr batch structure
Raw
ComBat
QC−RLSC
TIGER
SERRF
WiNN (QC−tuned)
WiNN (inferred segments)
WiNN
0.0
0.2
0.4
0.6
Weighted−PC R2
B BatchCorr replicate profiles
0.92
0.94
0.96
Replicate−profile Pearson r
C BatchCorr replicate agreement
0.875
0.900
0.925
0.950
Replicate ICC(A,1)
D Clinical plate structure
0.00
0.05
0.10
0.15
0.20
Weighted−PC R2
E Clinical replicate profiles
0.981
0.984
0.987
0.990
0.993
Replicate−profile Pearson r
F Clinical replicate agreement
0.980
0.985
0.990
Replicate ICC(A,1)


A Simulation
B MTBLS79
C Clinical FIA−MS
D BatchCorr Set 1
E Sacurine
F WaveICA
Batch weighted−PC R²
Raw
MAD
Drift
Batch
PQN


A Sacurine age
Covariate weighted−PC R² (%)
6
4
2
WiNN
QC−RLSC
SERRF
WiNN (QC−tuned)
WiNN (inferred segments)
ComBat
TIGER
Raw
0
5
10
15
20
Batch weighted−PC R² (%)


B Sacurine BMI
Covariate weighted−PC R² (%)
4
3
2
1
WiNN
WiNN (inferred segments)
WiNN (QC−tuned)
TIGER
SERRF
QC−RLSC
ComBat
Raw
0
5
10
15
20
Batch weighted−PC R² (%)


C Sacurine gender
Covariate weighted−PC R² (%)
10.0
7.5
5.0
2.5
WiNN (QC−tuned)
WiNN (inferred segments)
TIGER
WiNN
QC−RLSC
SERRF
ComBat
Raw
0
5
10
15
20
Batch weighted−PC R² (%)